NVMTS 2026

How Small Can a Ferroelectric Memory Bit Be?

Domain Switching and Scaling Limits in Fluorite-Structure Ferroelectrics

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MEMORY DENSITY

Ferroelectric domains in fluorite-structure materials

How small can a switchable domain be?

Venkatesan, Prasanna, et al. "Direct Observation of Nanoscale Polarization Switching in HZO Ferroelectrics Using STEM-EBIC." *2025 IEEE International Electron Devices Meeting (IEDM)*. IEEE, 2025. 10.1109/IEDM50572.2025.11353770

Collaborator: Chris Regan (UCLA)

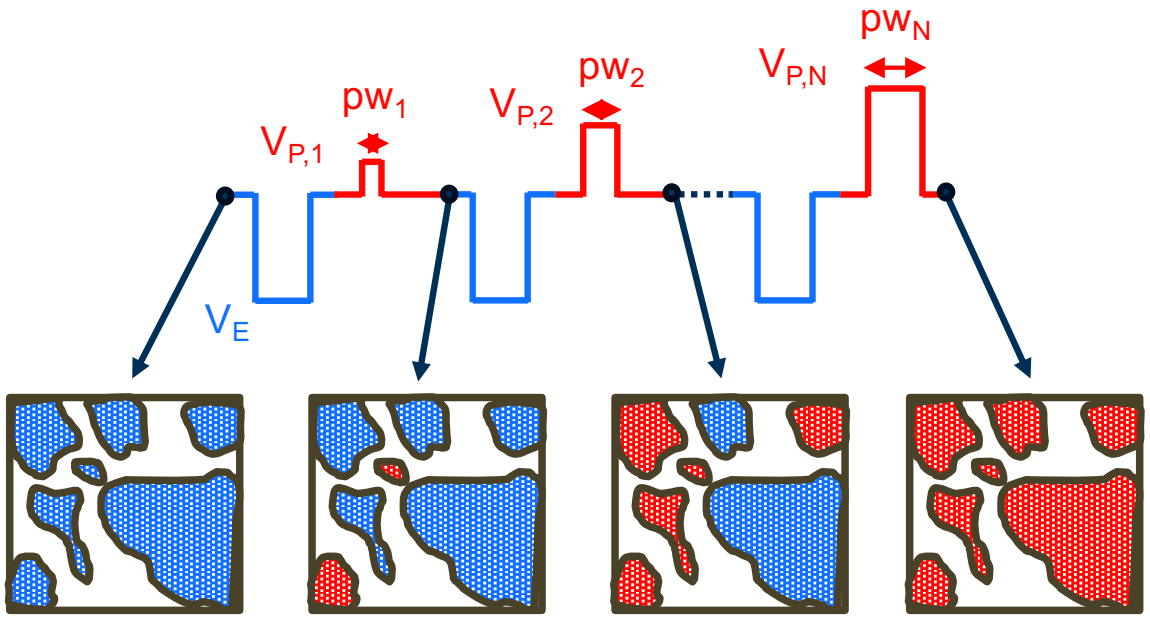
How small can a domain become?

N. Afroze et al., “Sub-nm² Ferroelectric Domains via Charged 180° Walls in ZrO_2,” *Advanced Materials* (2026), e75144. DOI: 10.1002/adma.75144
Preprint: https://arxiv.org/abs/2507.18920

Collaborators: **Rohan Mishra (WashU), Jun Hee Lee (UNIST, South Korea), Christopher Nelson (ORNL), Andrew Lupini (ORNL)**

DEFINITION

Electrically switchable domains



Electrical domains are defined as an ensemble of contiguous ferroelectric dipoles that switch at the same external field and pulse width (V, pw)

MEASUREMENT

STEM EBIC – Scanning Transmission Electron Microscopy – Electron Beam-Induced Current

Process Flow

Anneal at 700 ℃

↑

Sputter TE

↑

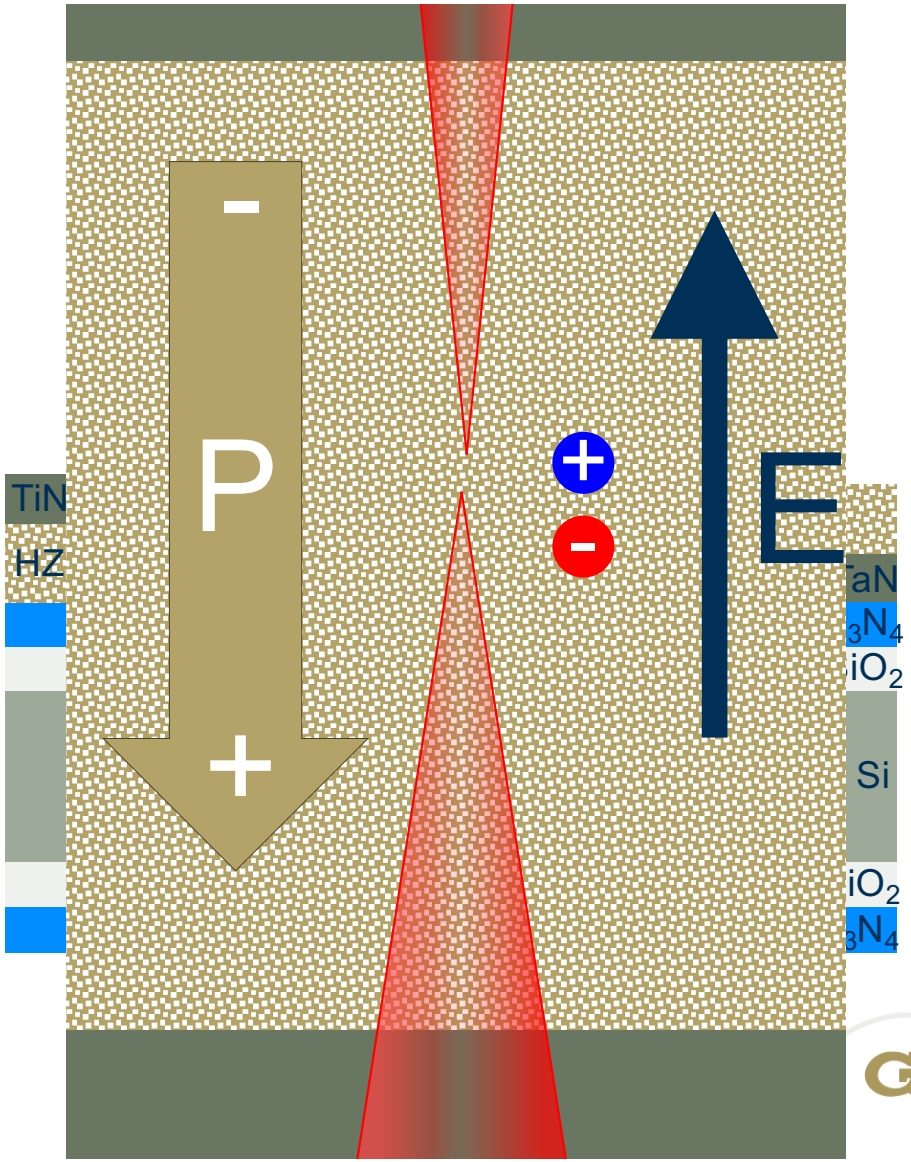
ALD HZO

↑

Sputter BE

↑

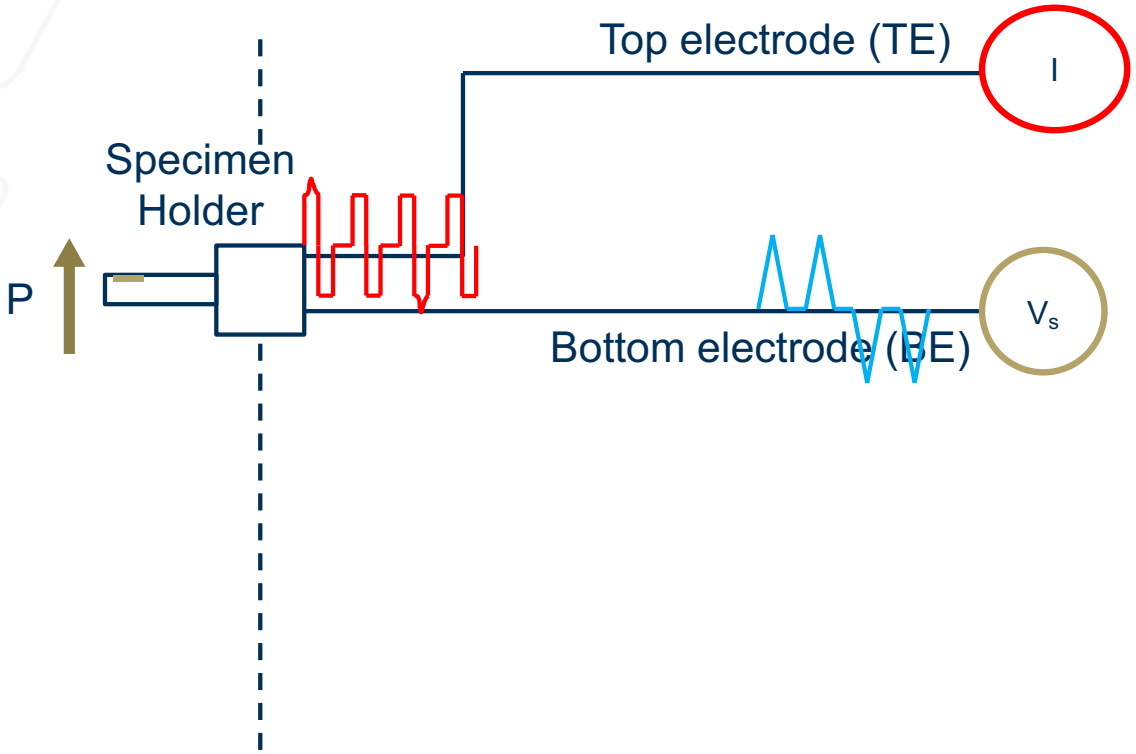
SiN/Si Substrate

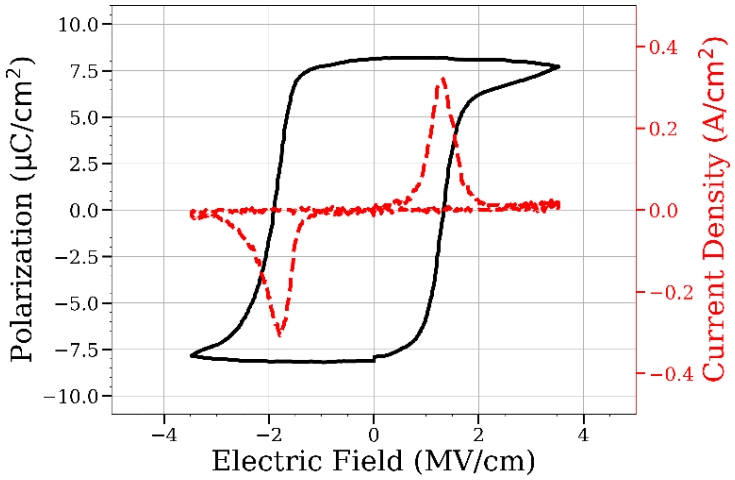


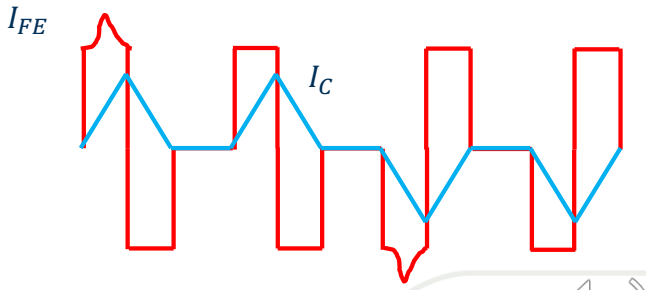
Collaborator: Chris Regan (UCLA)

MEASUREMENT

In-situ polarization measurement

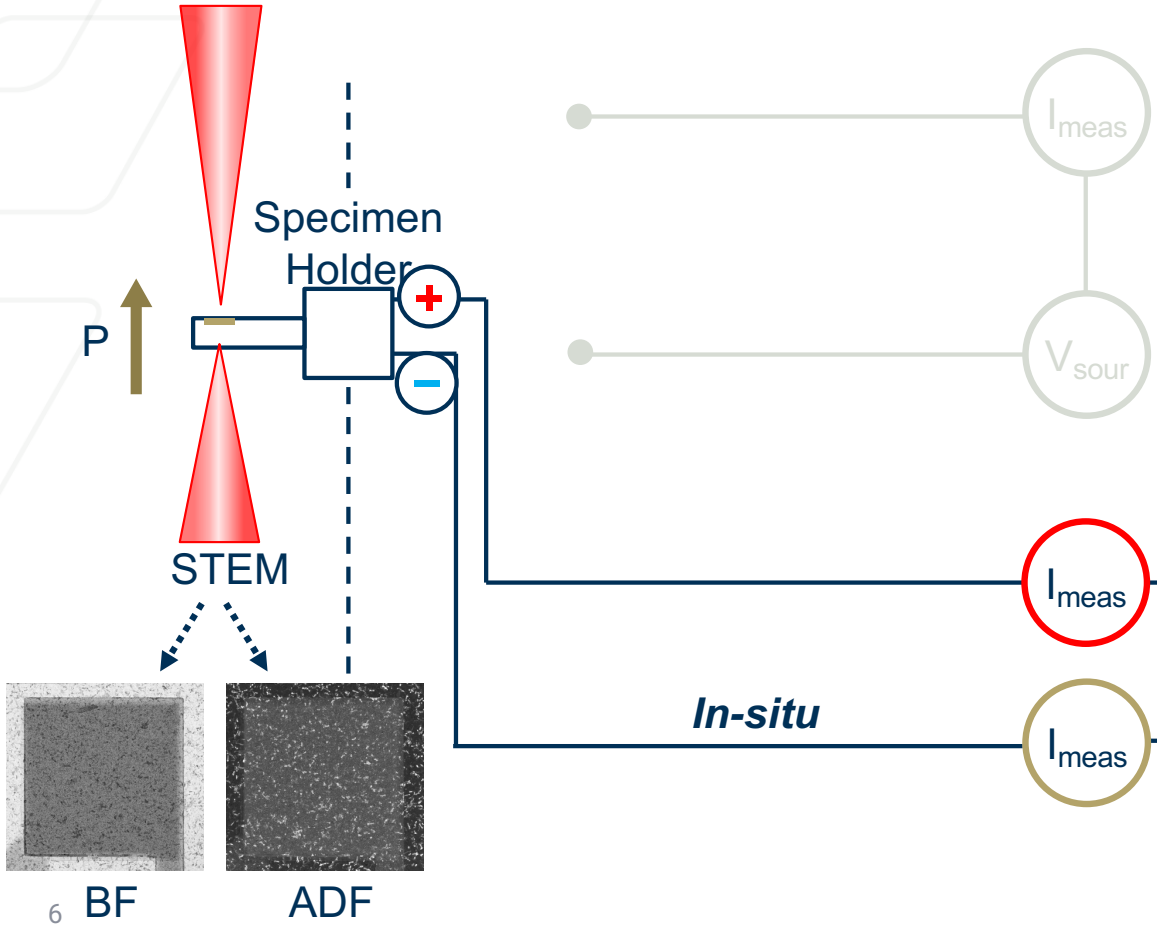


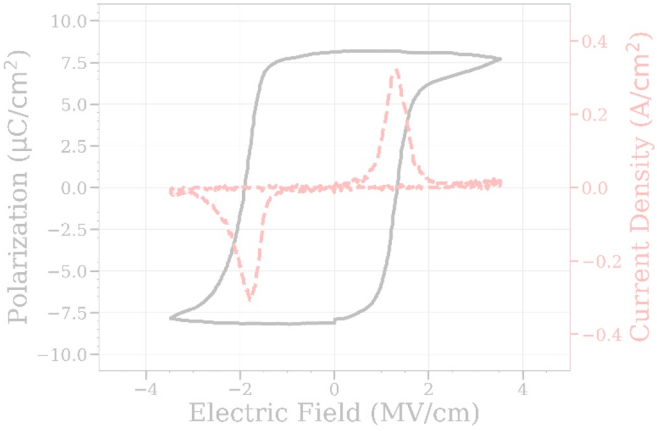




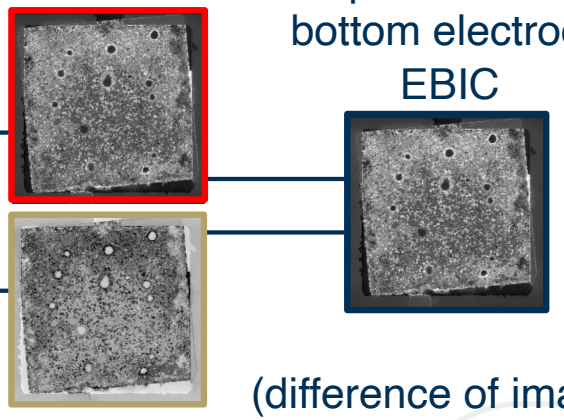
MEASUREMENT

EBIC measurement





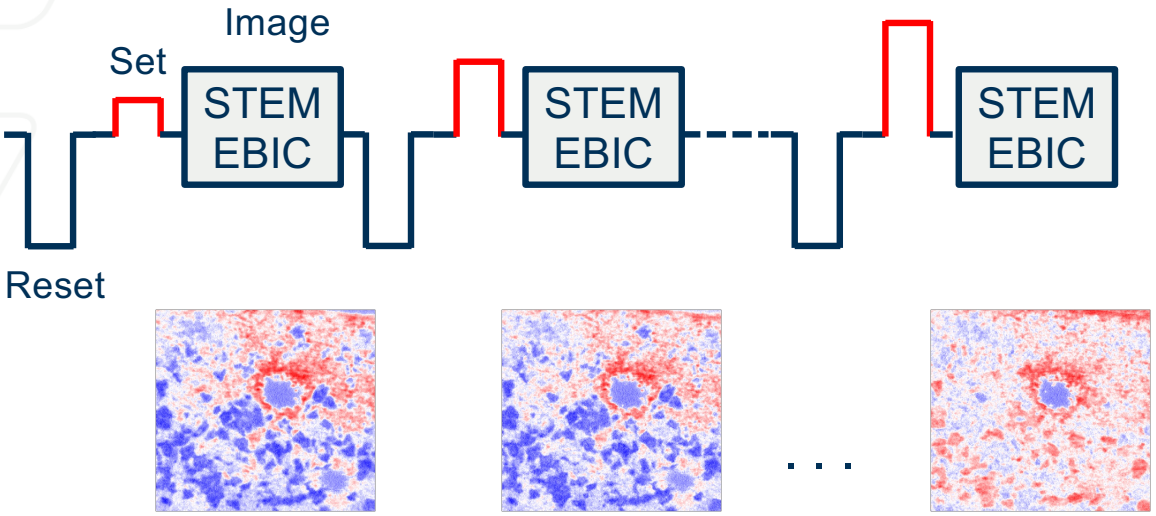
top electrode -
bottom electrode
EBIC

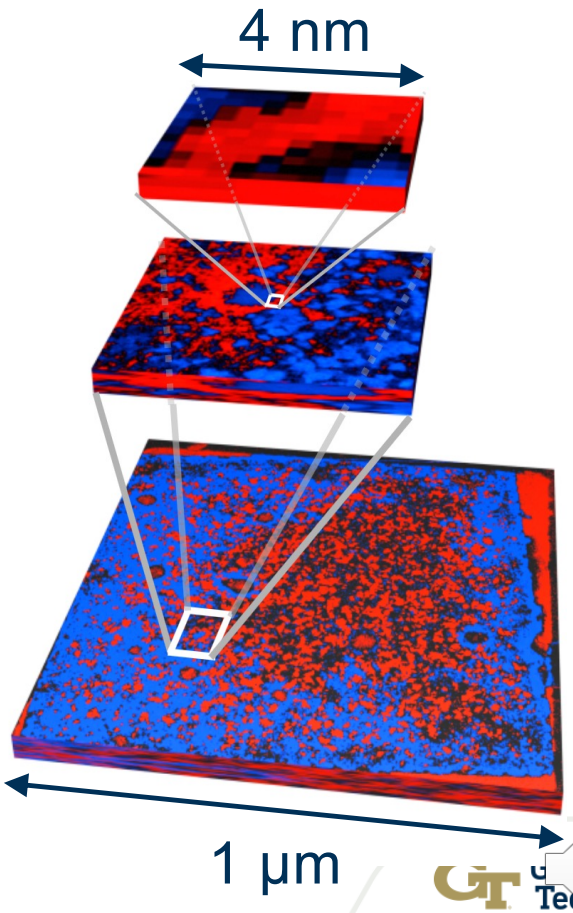


(difference of images
acquired simultaneously)

MEASUREMENT

Pulse sequence

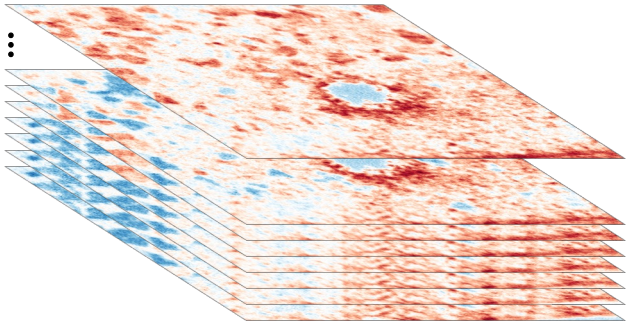




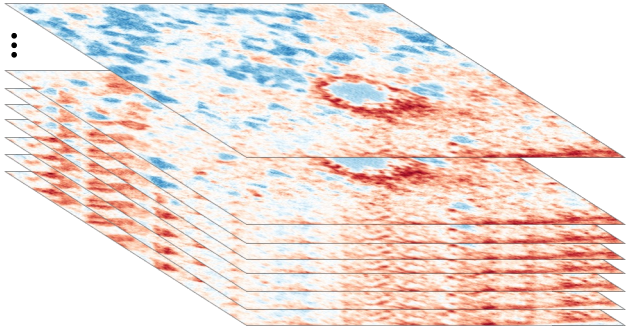
MEASUREMENT

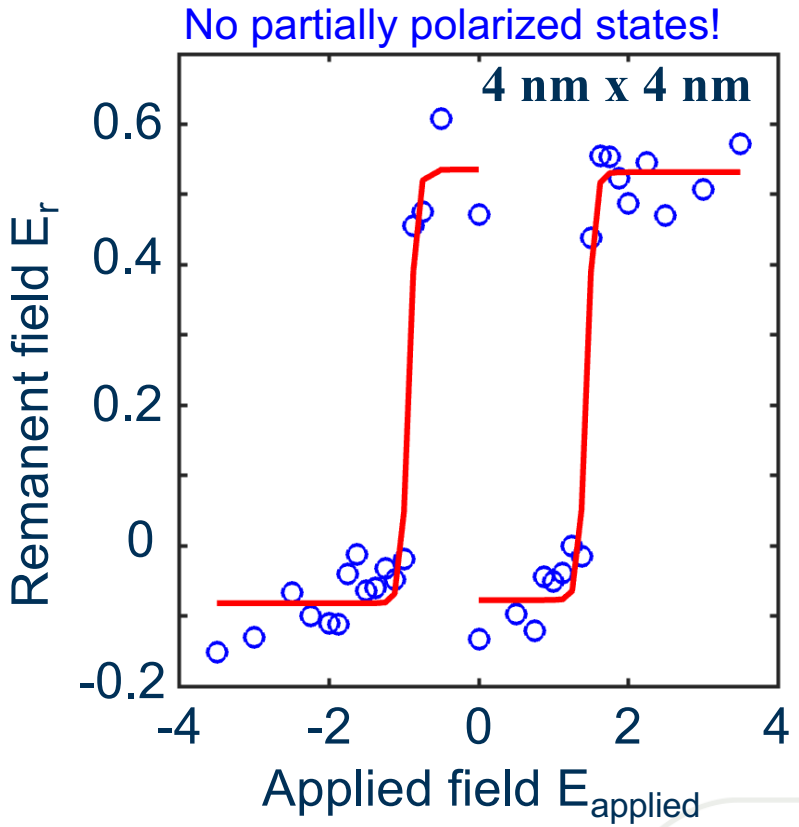
EBIC Polarization-Field Characteristics

PUNDp
$P_{\uparrow} \rightarrow P_{\downarrow}$
17 maps



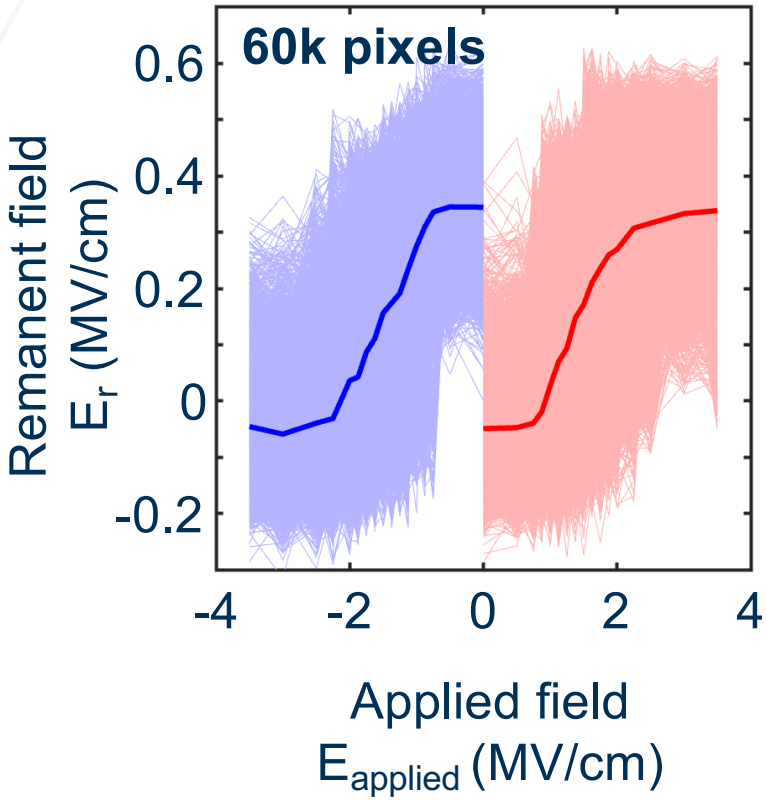
NDPUn
$P_{\downarrow} \rightarrow P_{\uparrow}$
17 maps

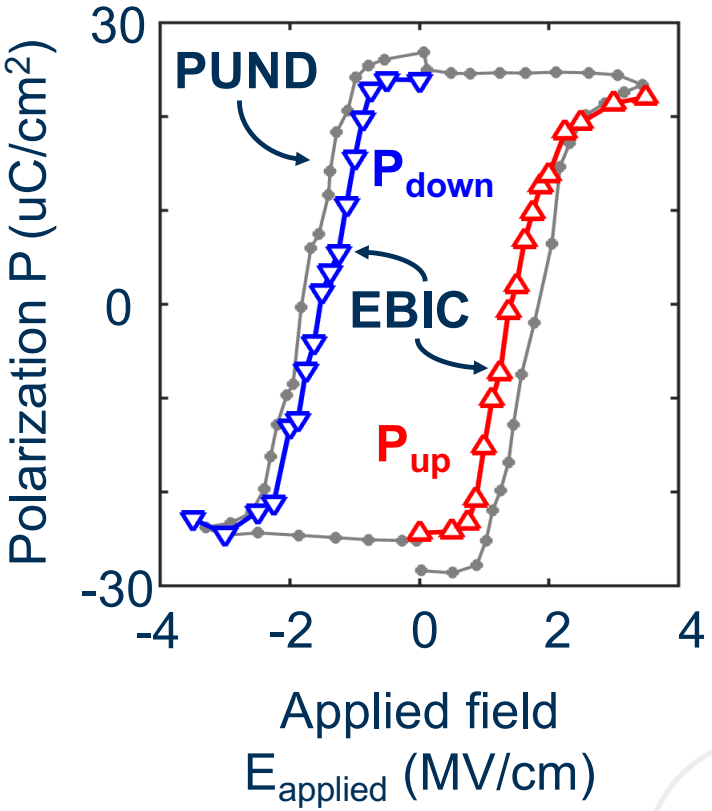




MEASUREMENT

Device-level Polarization-Field Characteristics

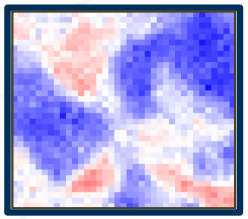




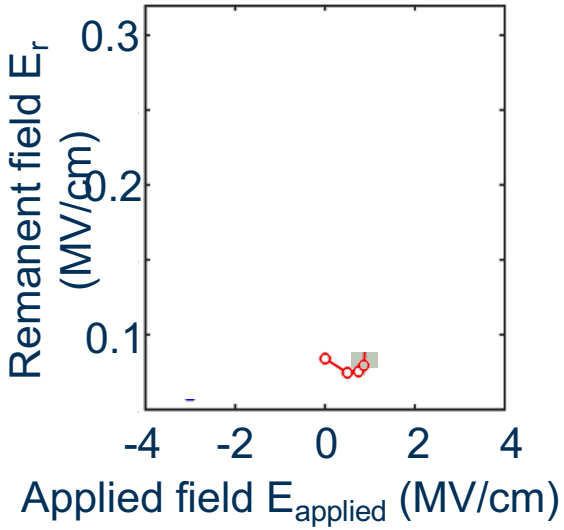
Mapping local switching events

Uncorrelated single domain switching

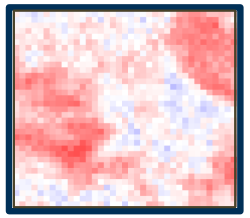
E_{app} = 0 MV/cm



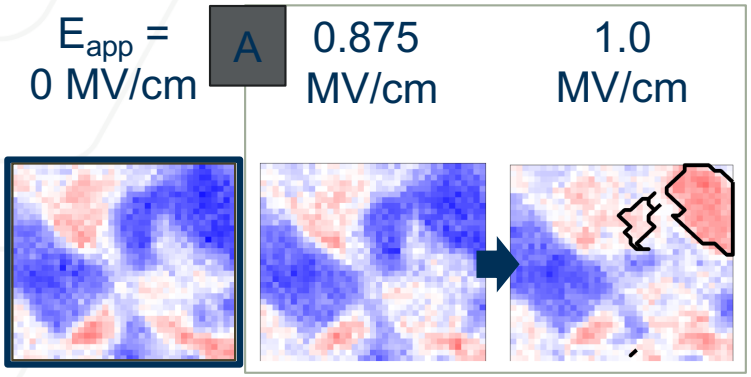
47.5 nm x 47.5 nm

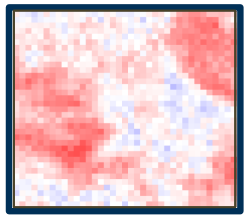


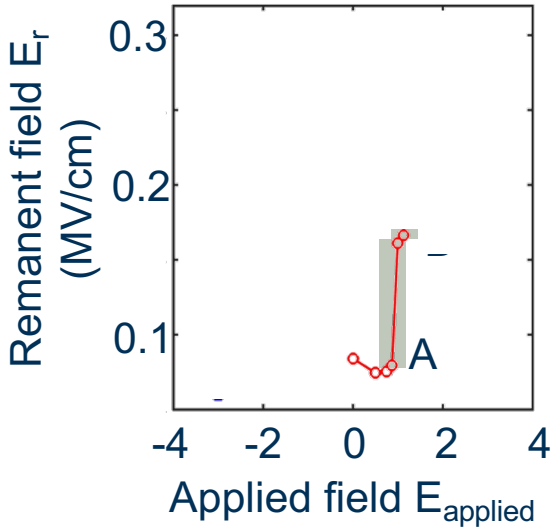
3.5 MV/cm



Uncorrelated single domain switching

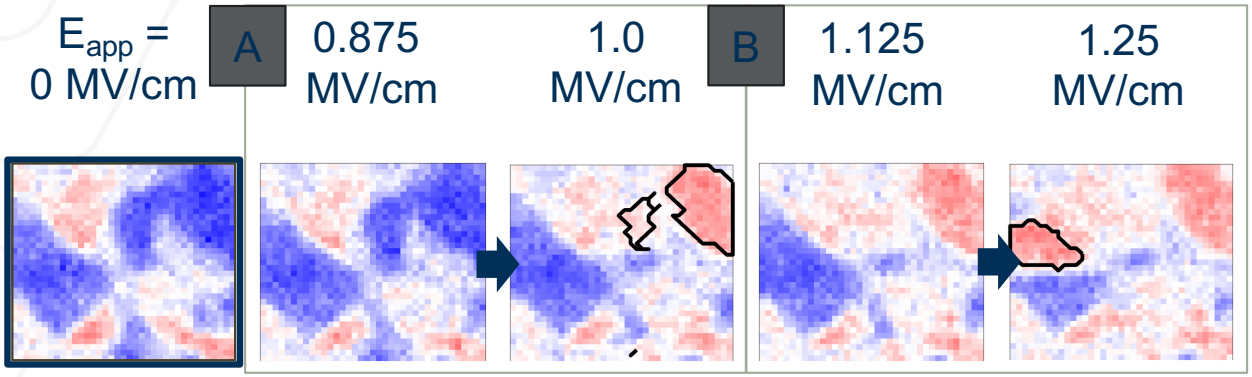


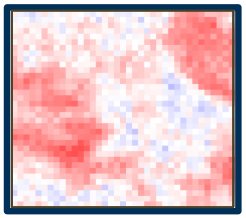


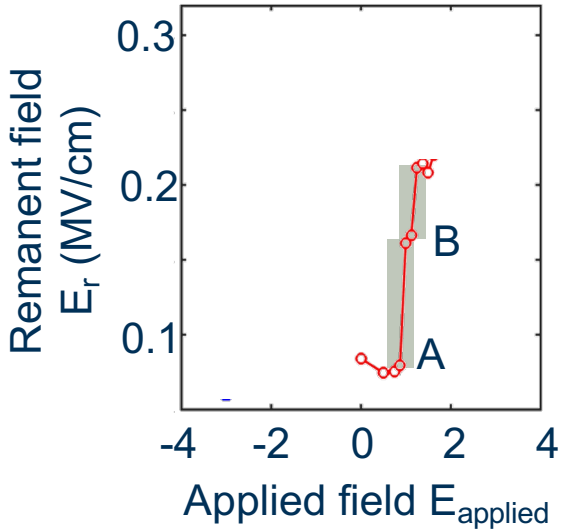


Mapping local switching events

Uncorrelated single domain switching

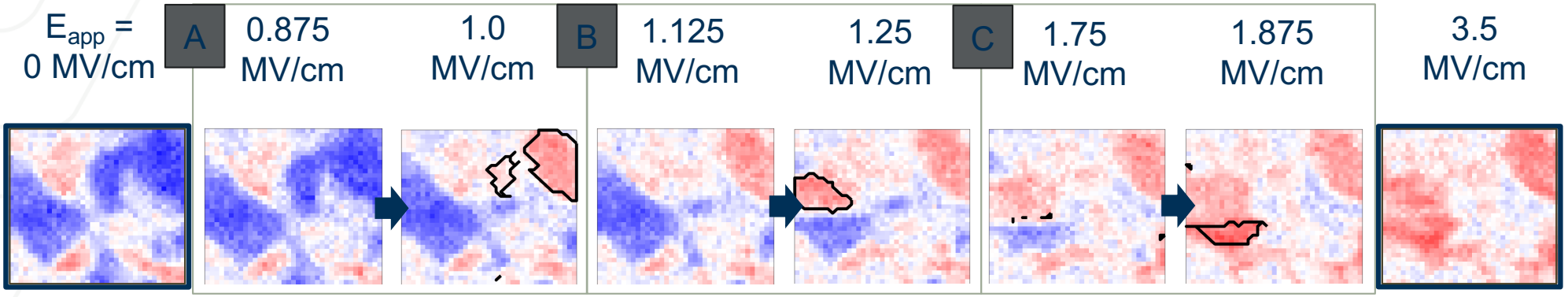


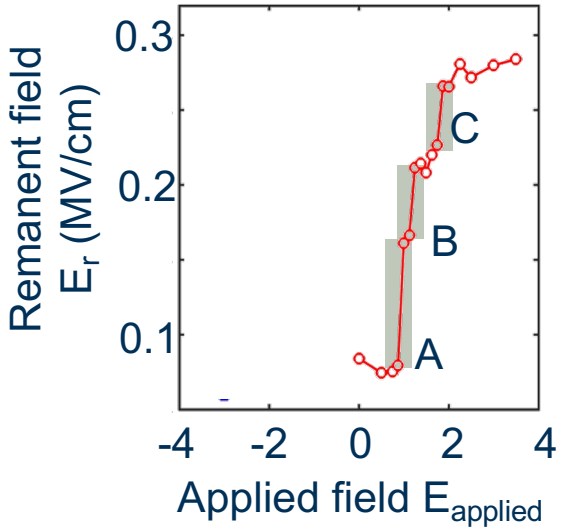




Mapping local switching events

Uncorrelated single domain switching

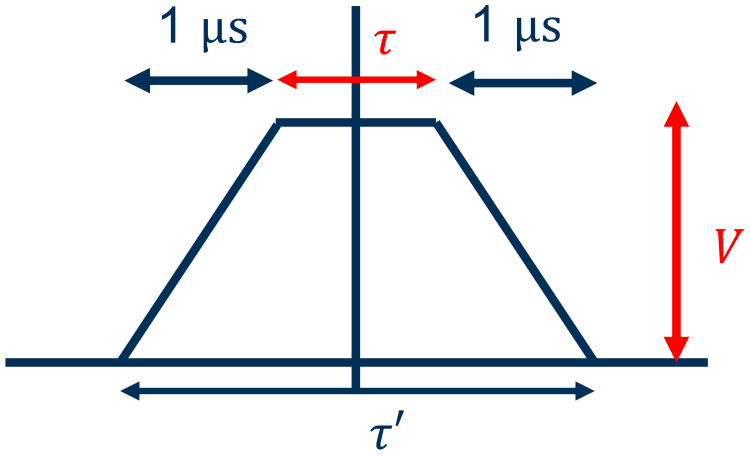


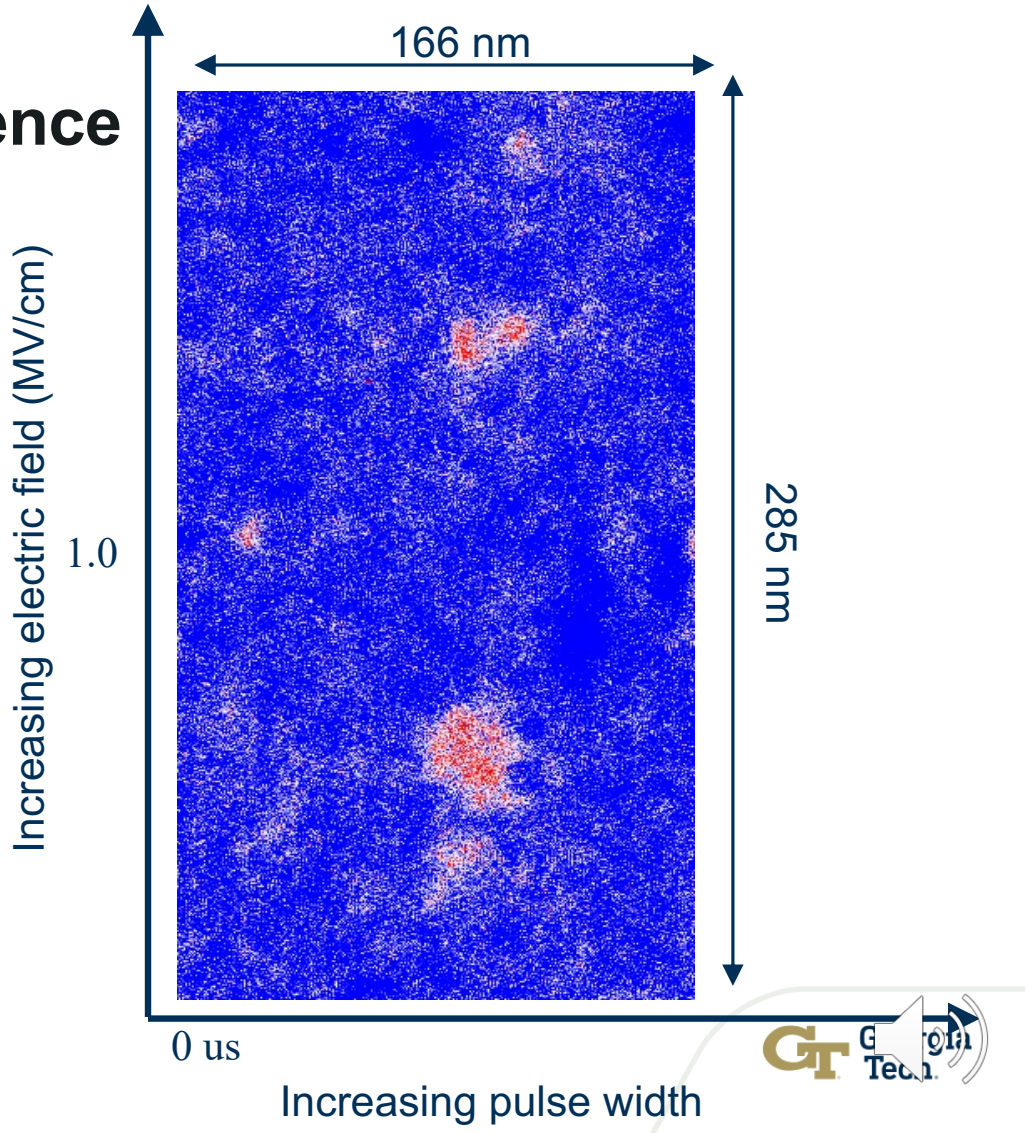


Field and pulse-width dependence

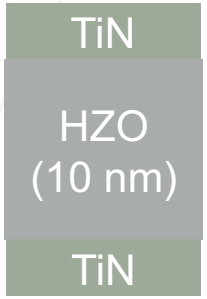


Set pulse

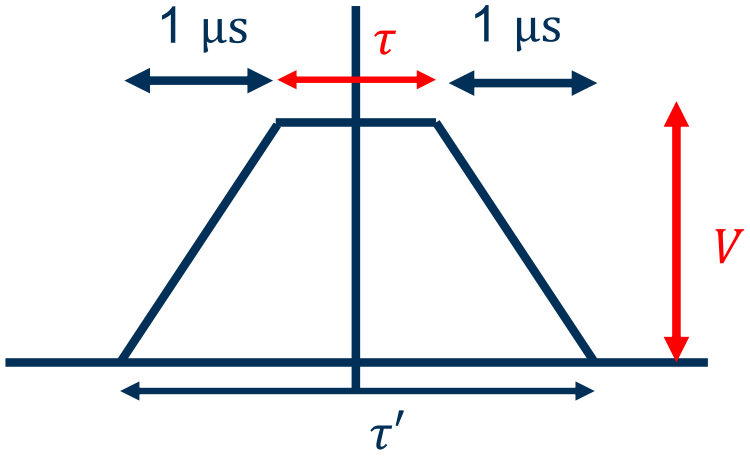


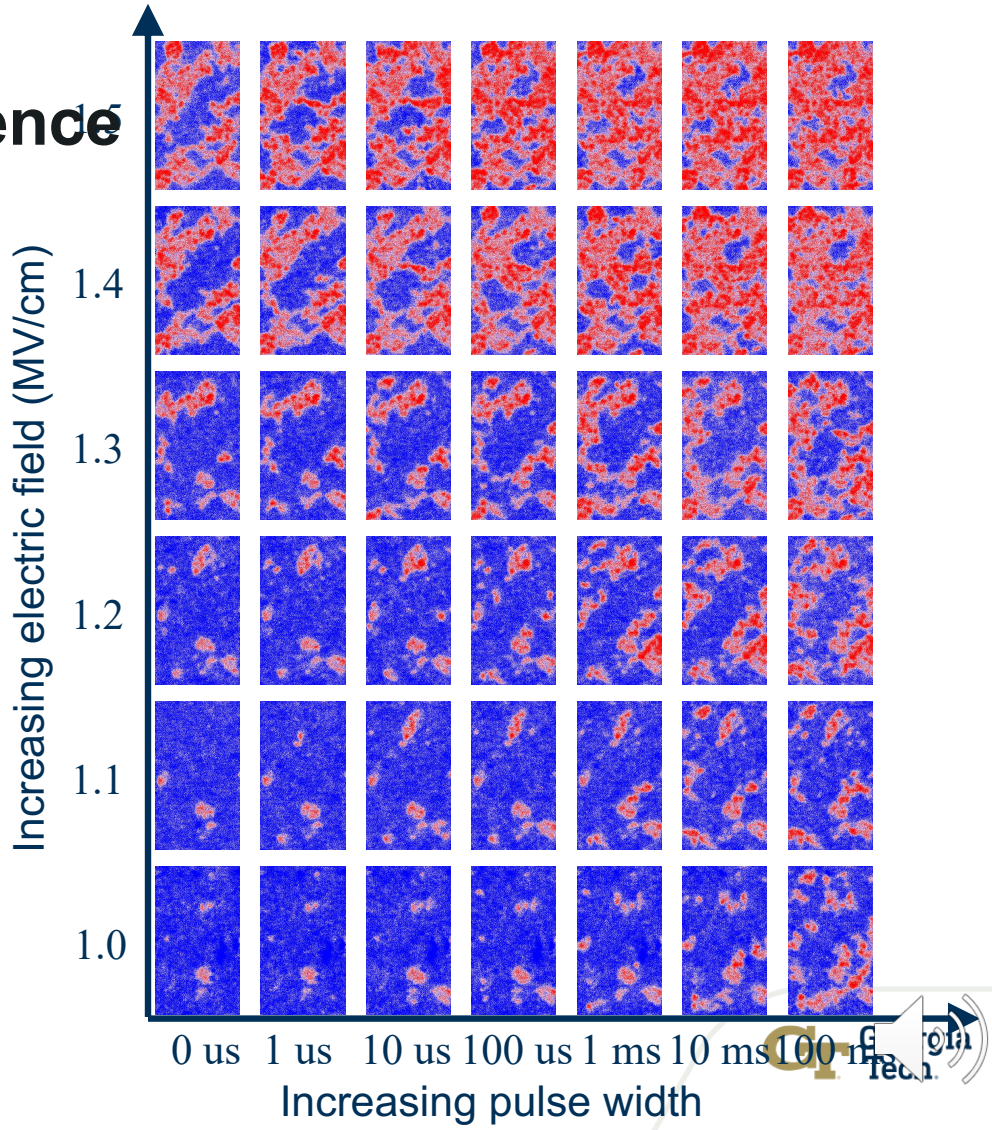


Field and pulse-width dependence



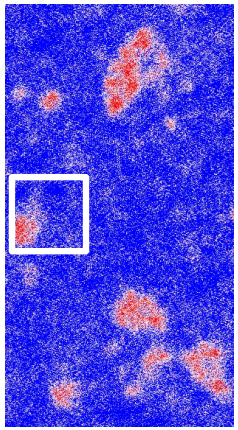
Set pulse





SWITCHING DYNAMICS

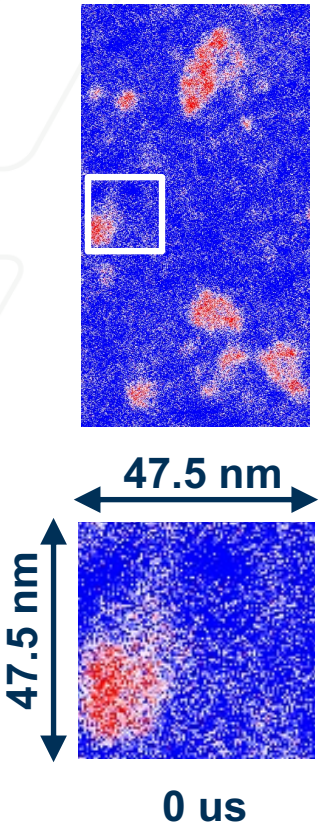
Constant field (1.2 MV/cm)



0 us

SWITCHING DYNAMICS

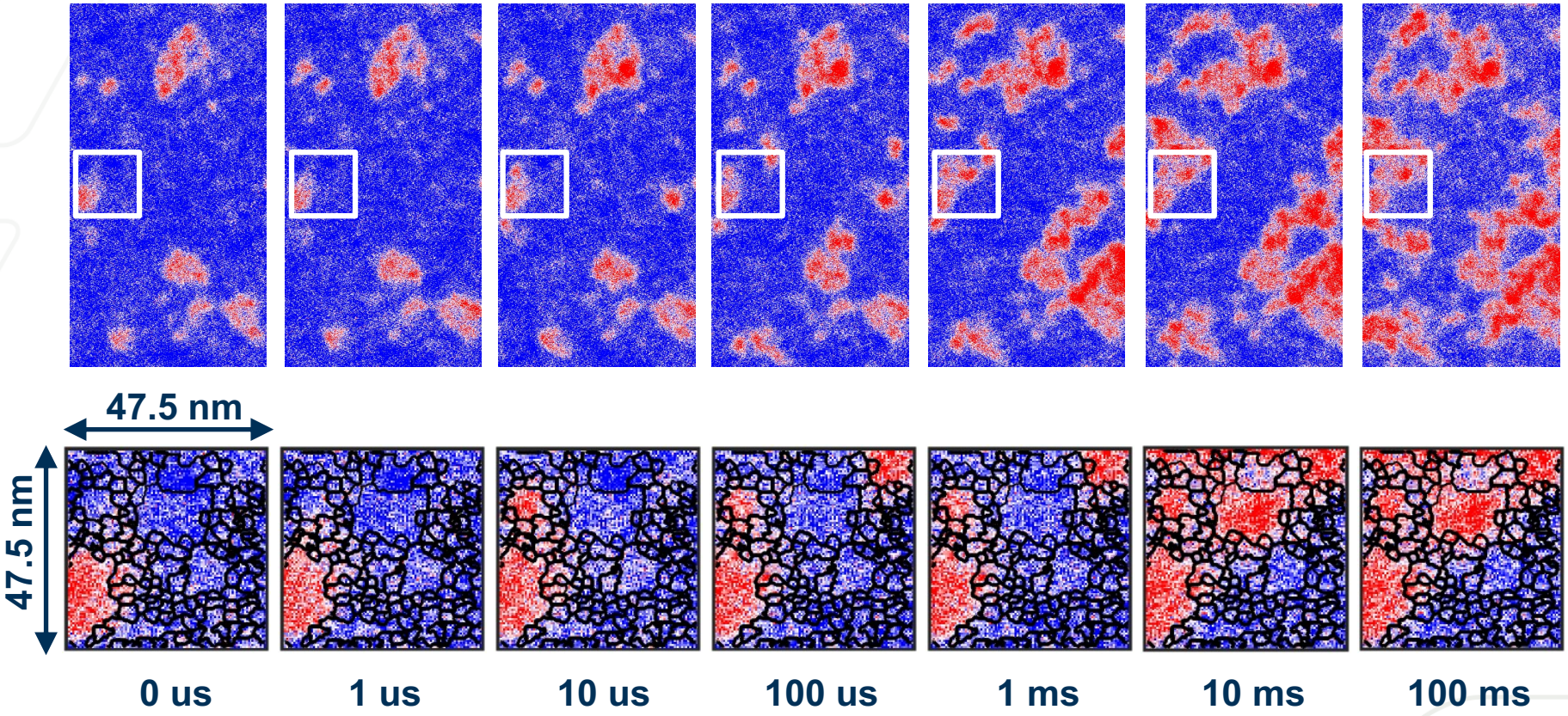
Constant field (1.2 MV/cm)



REDO recording

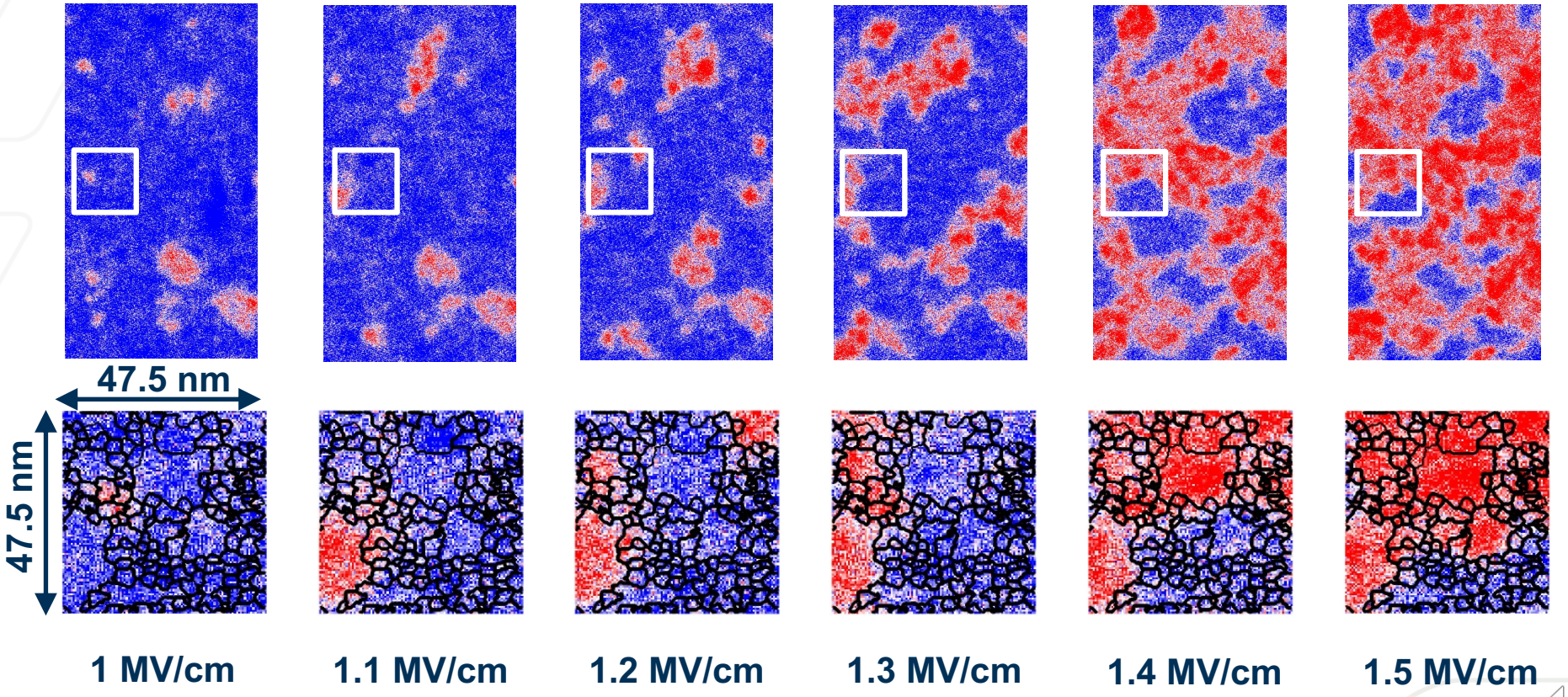
SWITCHING DYNAMICS

Constant field (1.2 MV/cm)



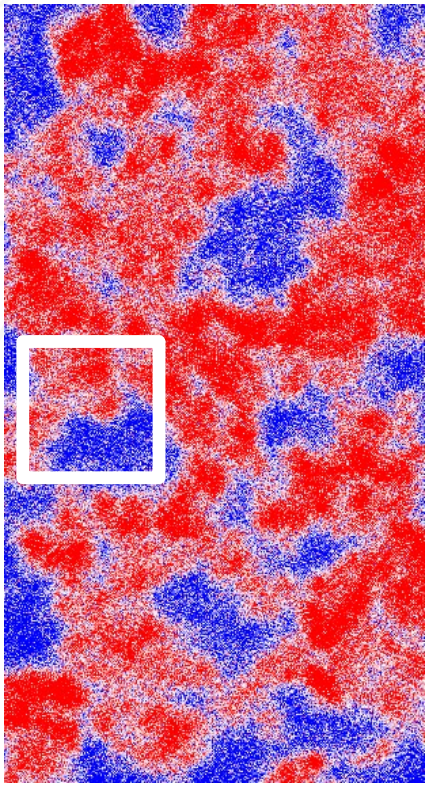
SWITCHING DYNAMICS

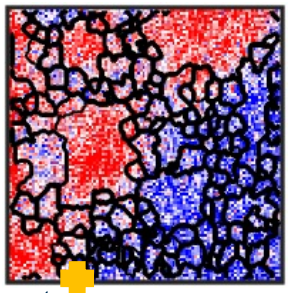
Constant pulse width (100 us)



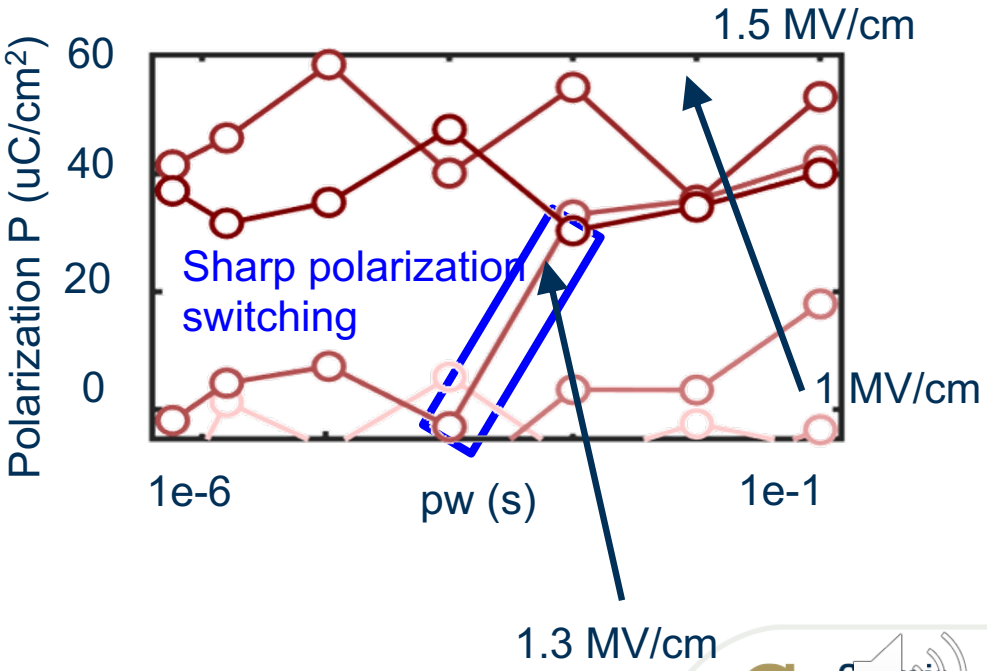
SWITCHING DYNAMICS

At different spatial length scales



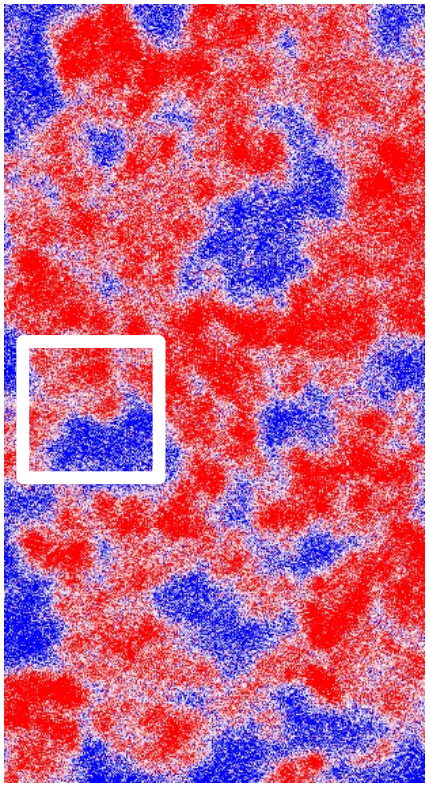


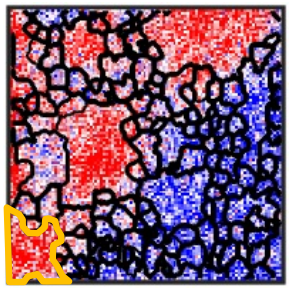
Pixel A:
0.475 nm x 0.475 nm



SWITCHING DYNAMICS

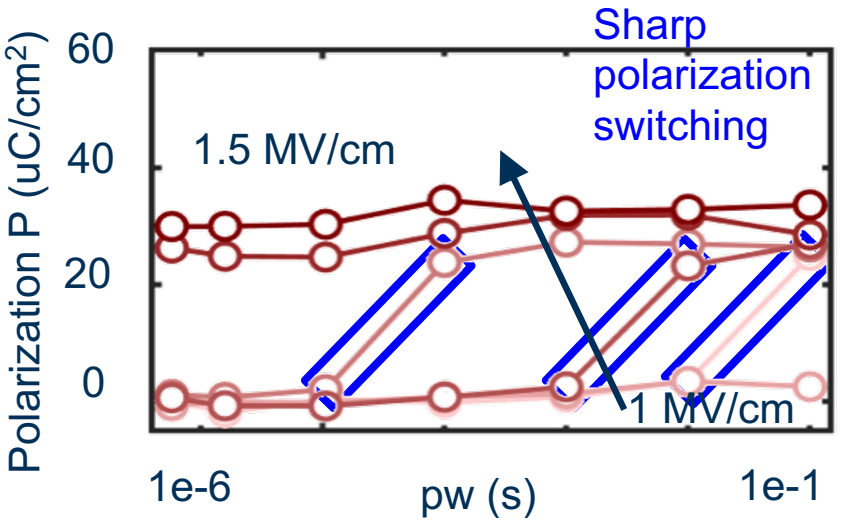
At different spatial length scales





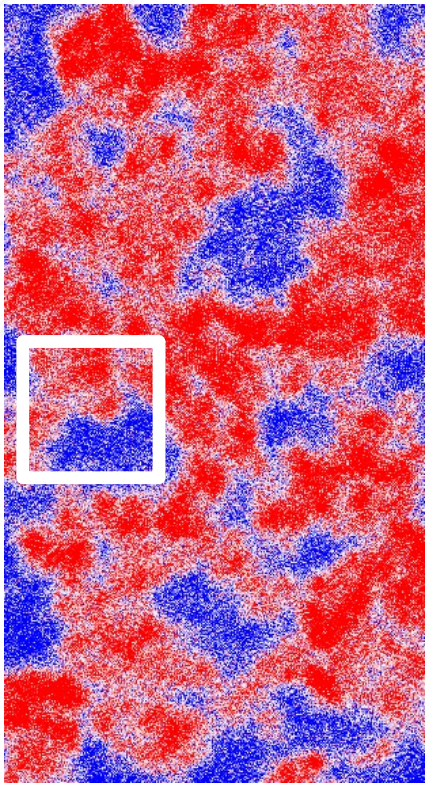
Domain #B
(~7 nm x 7 nm)

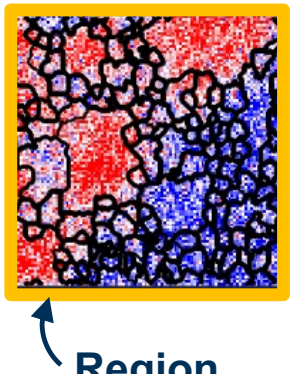
No partially polarized states!



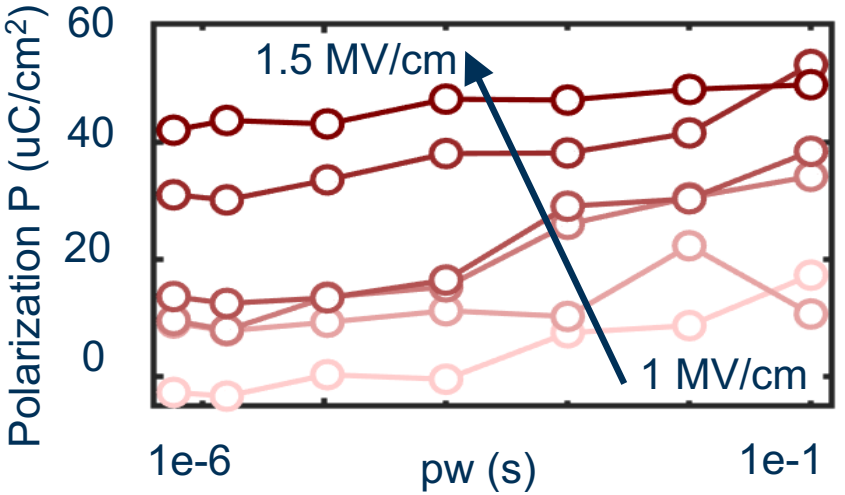
SWITCHING DYNAMICS

At different spatial length scales

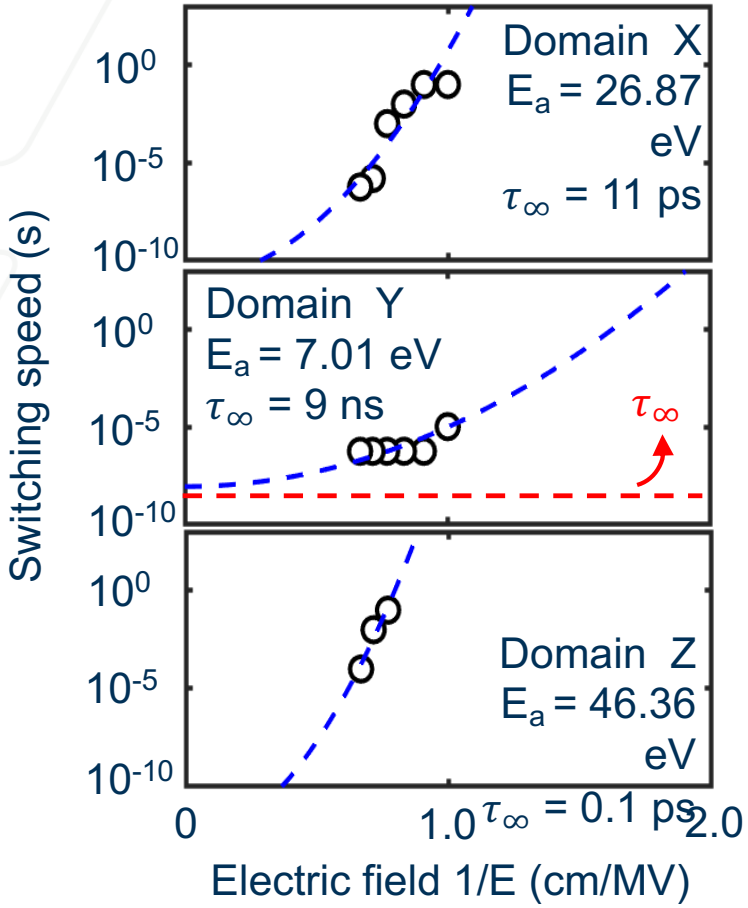




Region
47.5 nm x 47.5 nm



Field dependence of domain switching time



Switching time of a domain,

$$\tau = \tau_\infty exp\left(\left[\frac{E_a}{E}\right]^2\right)$$

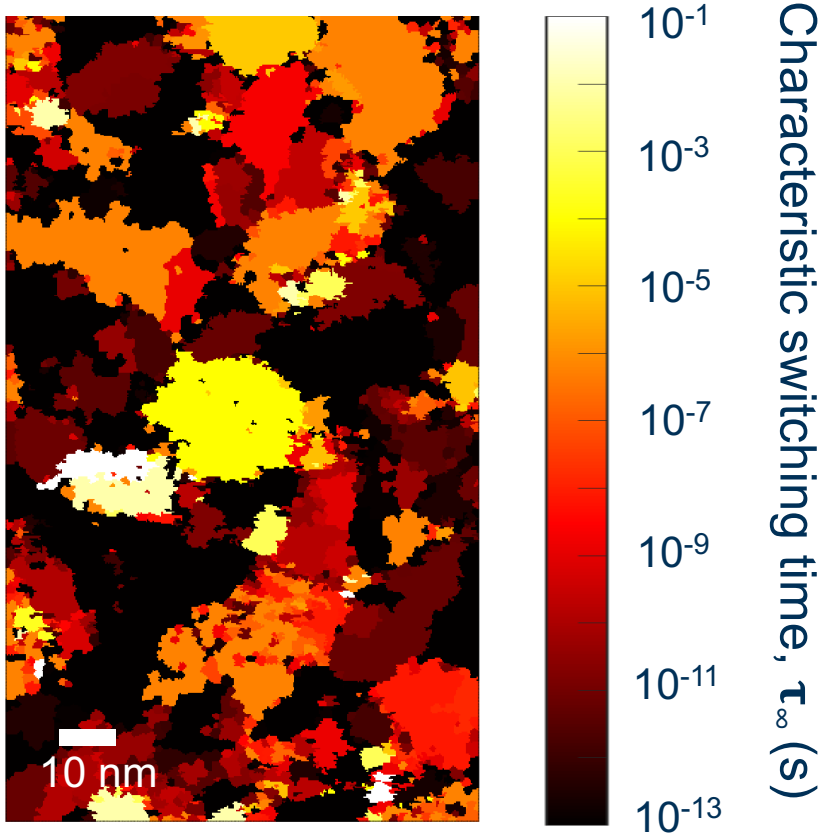
Activation field

Characteristic switching time

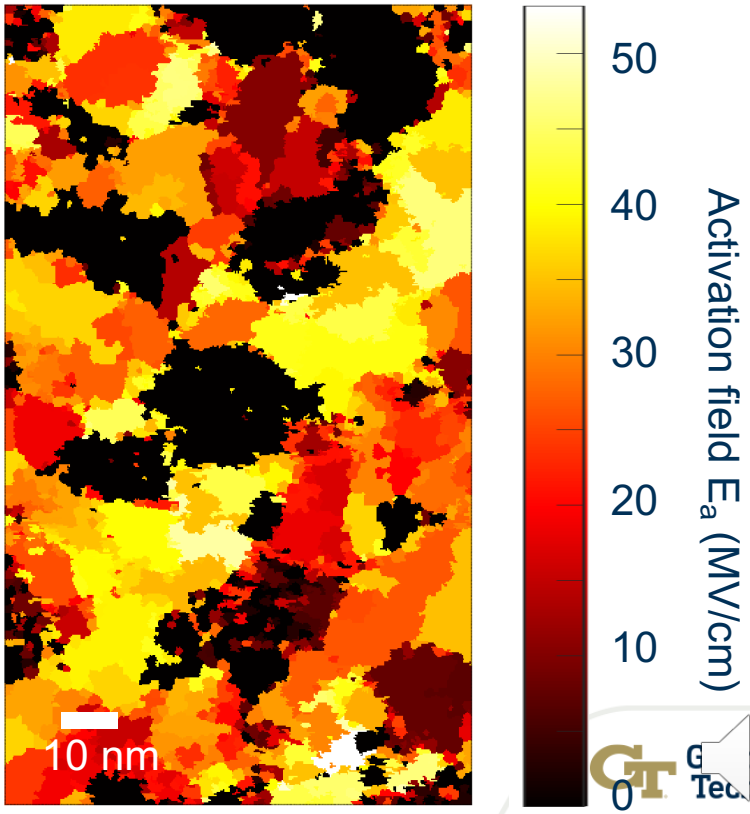
Applied electric field

Spatial variation in domain switching kinetics

Characteristic switching time, τ_∞



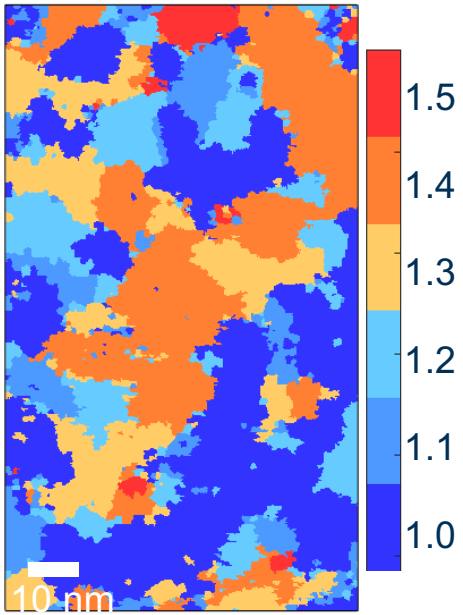
Activation energy, E_a



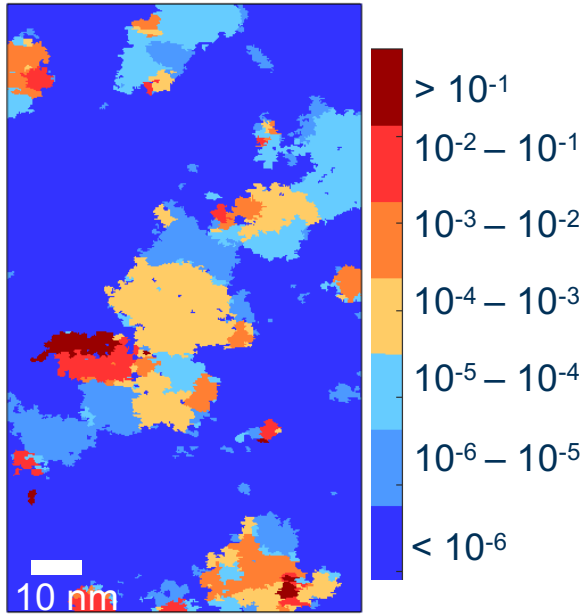
SWITCHING DYNAMICS

Spatial variation in switching properties

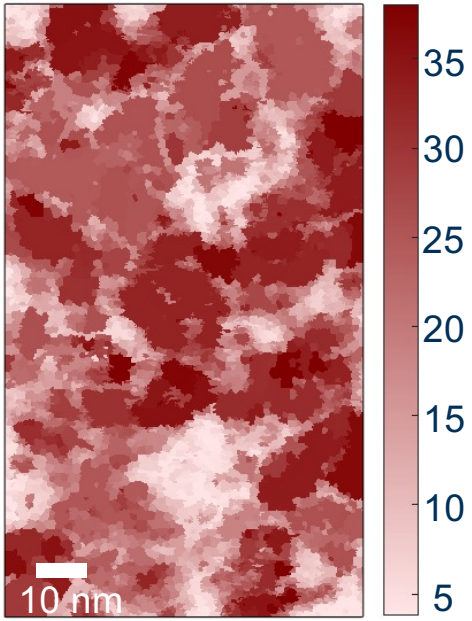
Minimum switching field



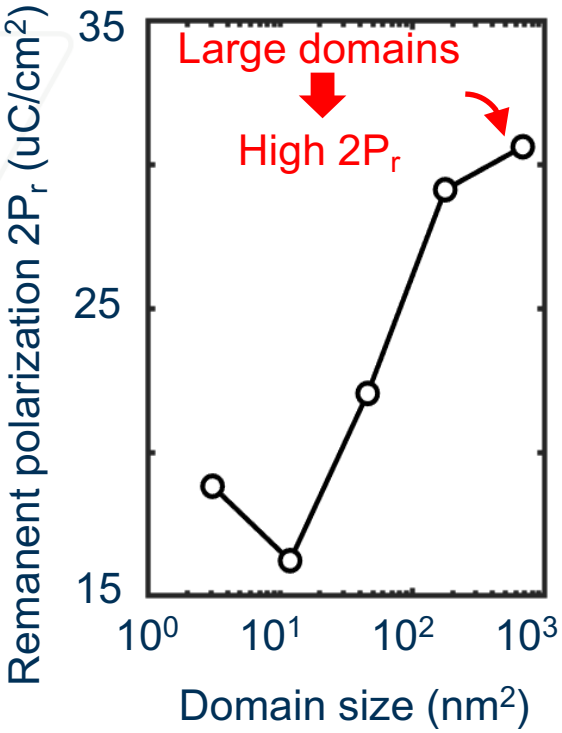
Switching time at 1.5 MV/cm (s)

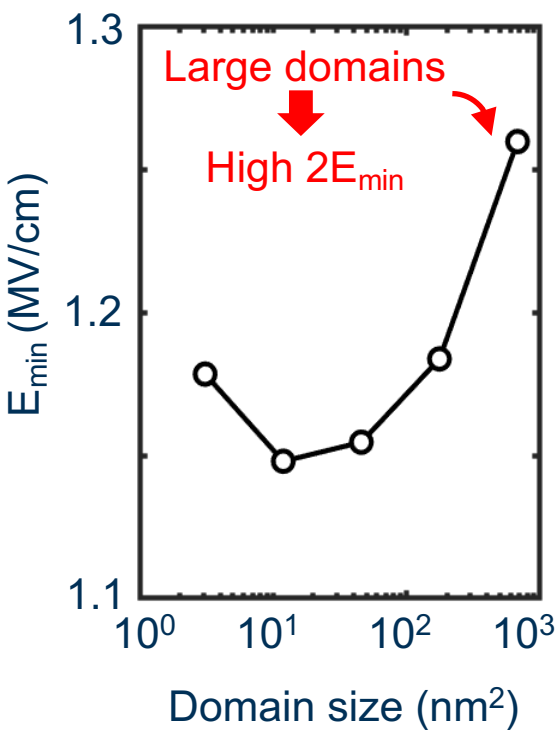


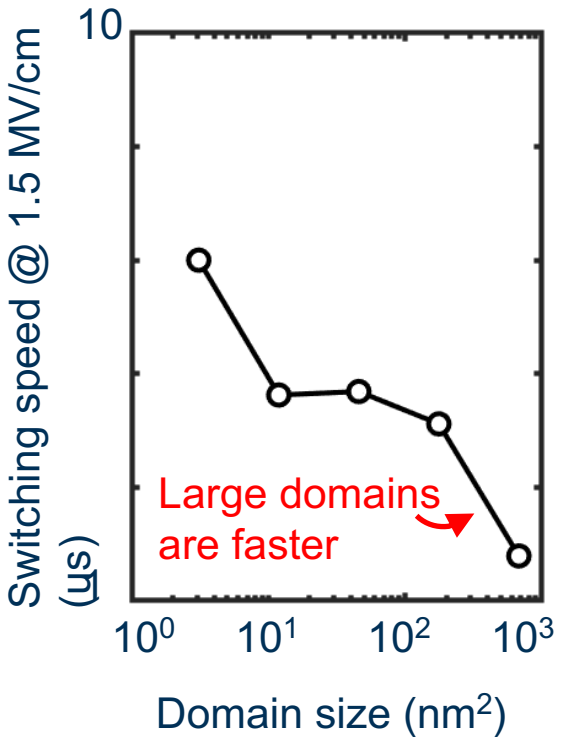
Remanent polarization 2Pr (μC/cm²)



INSIGHTS INTO DOMAIN SIZE DEPENDENCE

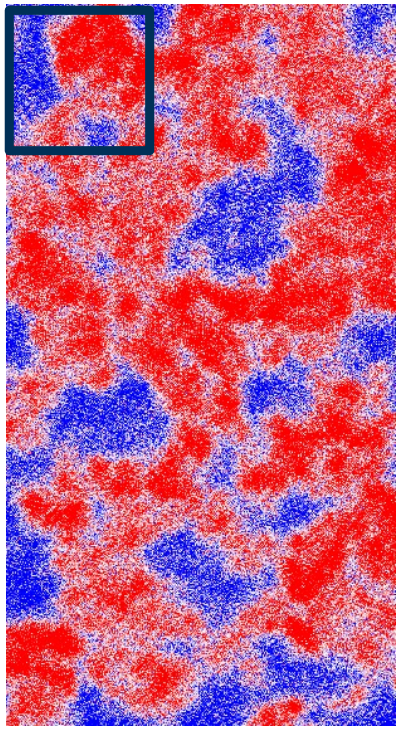


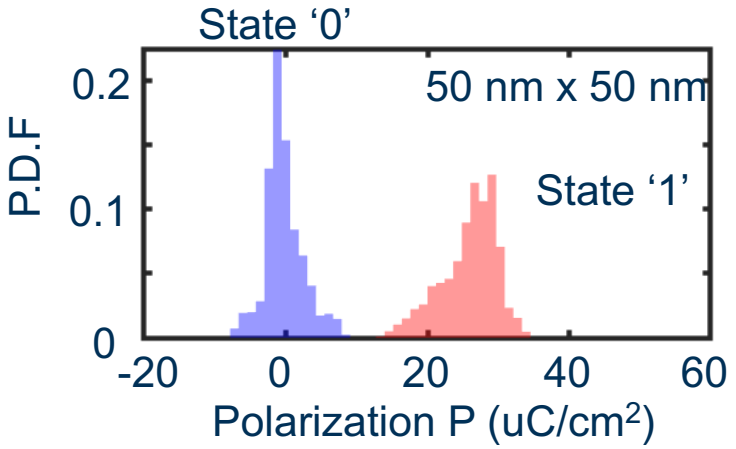




Variation

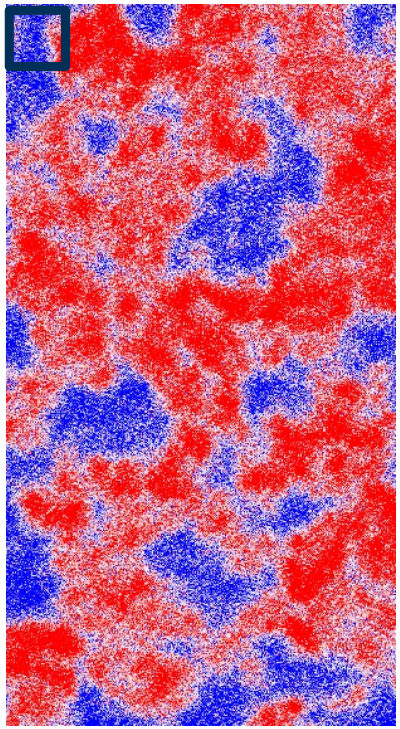
Device-to-device Variation in Scaled Ferroelectrics

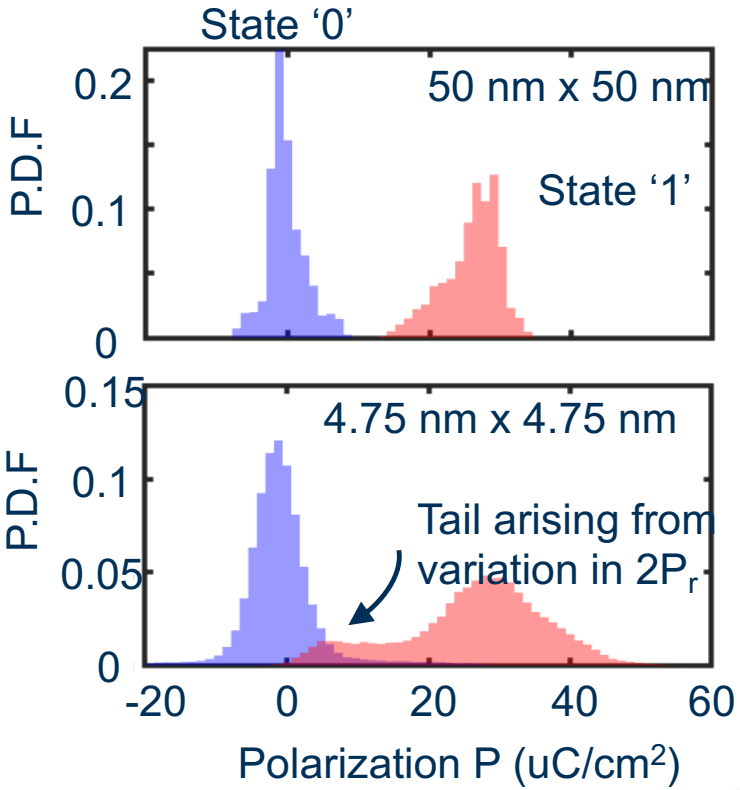




Scaled devices show a tail leading to increased bit error rate

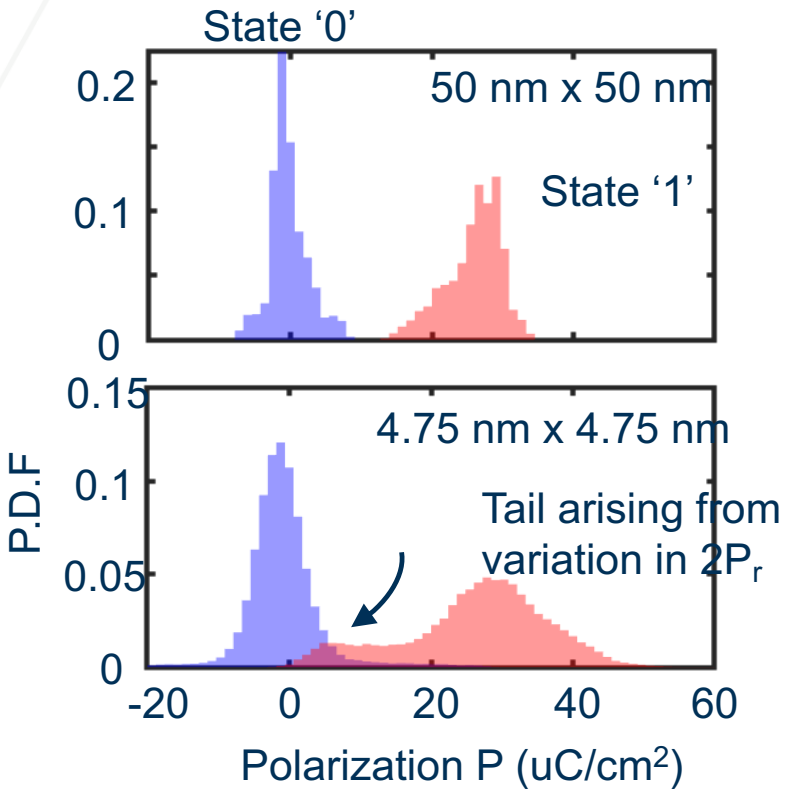
Device-to-device Variation in Scaled Ferroelectrics

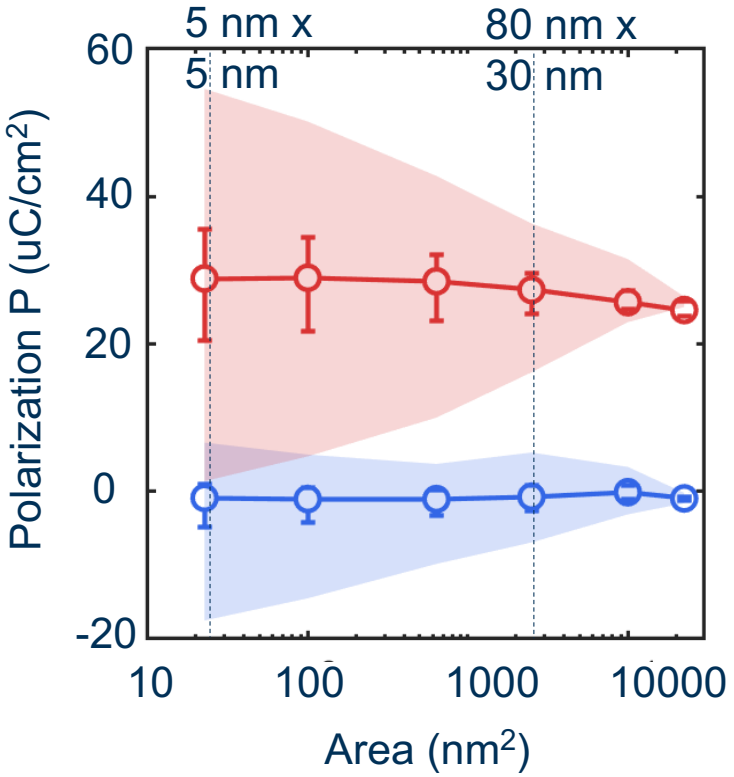




Scaled devices show a tail leading to increased bit error rate

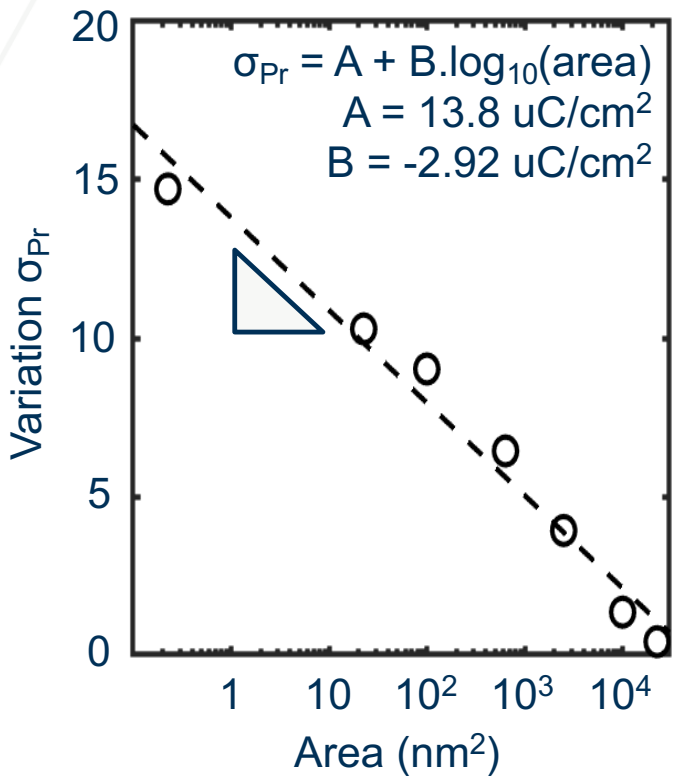
Device-to-device Variation in Scaled Ferroelectrics

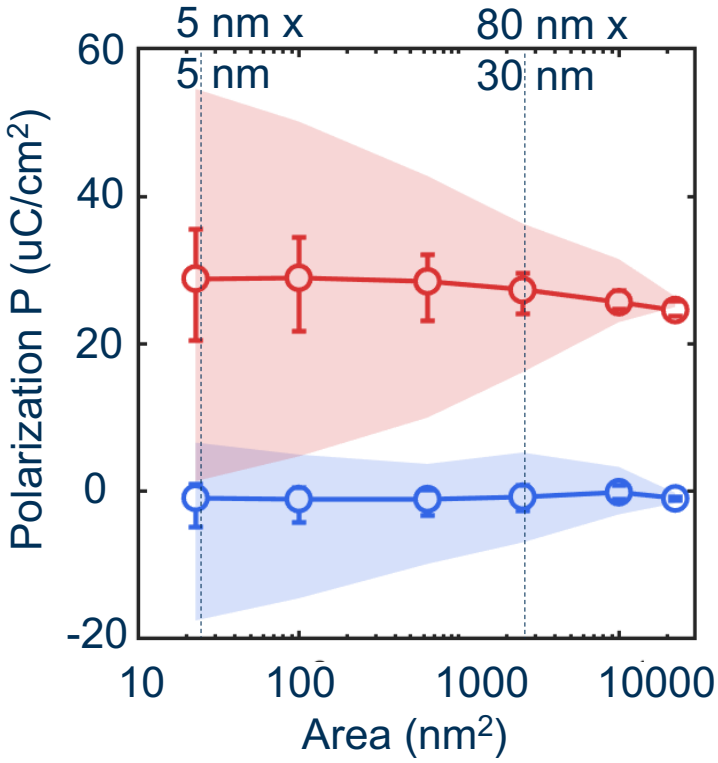




Variation remains minimal at DRAM and FEFET dimensions

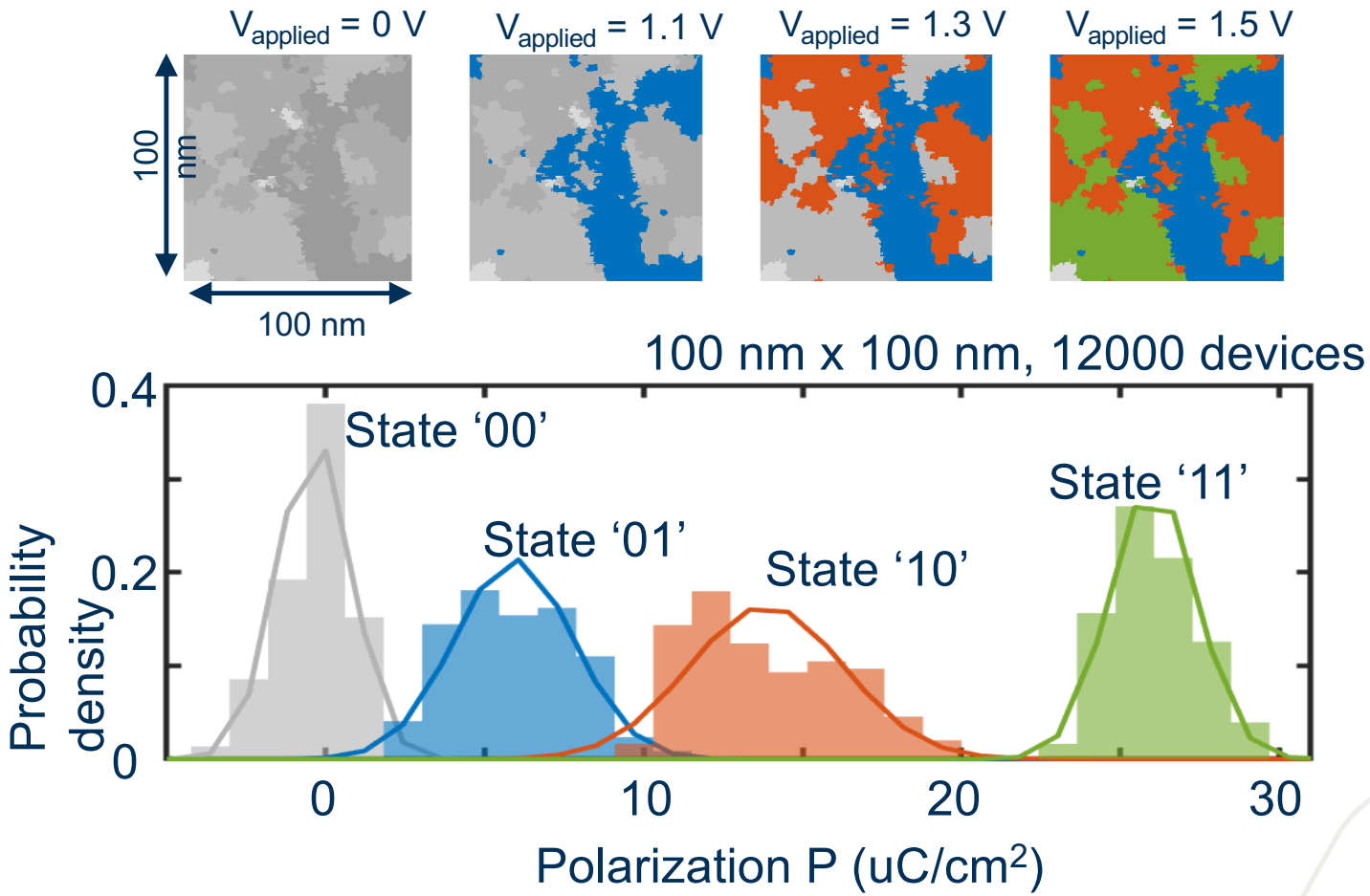
Device-to-device Variation in Scaled Ferroelectrics





The variation in P_r increases linearly with every decade reduction in area.

Ensemble domain switching for multi-bit operation



MEMORY DENSITY

Domain switching and memory density

How small can a switchable domain be?

Venkatesan, Prasanna, et al. "Direct Observation of Nanoscale Polarization Switching in HZO Ferroelectrics Using STEM-EBIC." *2025 IEEE International Electron Devices Meeting (IEDM)*. IEEE, 2025. 10.1109/IEDM50572.2025.11353770

Collaborator: Chris Regan (UCLA)

How small can a domain become?

N. Afroze et al., “Sub-nm² Ferroelectric Domains via Charged 180° Walls in ZrO_2,” *Advanced Materials* (2026), e75144. DOI: 10.1002/adma.75144
Preprint: https://arxiv.org/abs/2507.18920

Collaborators: **Rohan Mishra (WashU), Jun Hee Lee (UNIST, South Korea), Christopher Nelson (ORNL), Andrew Lupini (ORNL)**

FLUORITE SCALING

Confinement perpendicular to the polar axis

The fluorite lattice alternates polar and spacer sheets

Polar sheet width

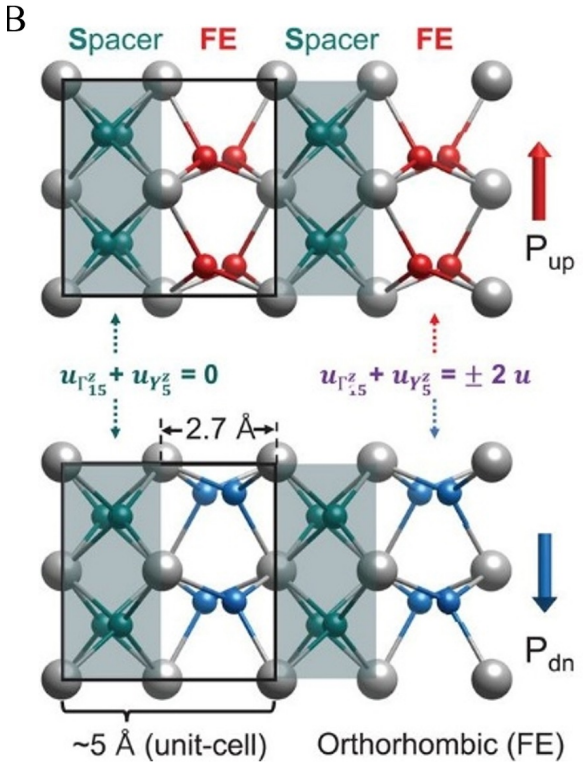
2.7 Å

Half of one fluorite unit cell

Sheet-to-sheet repeat

≈5 Å

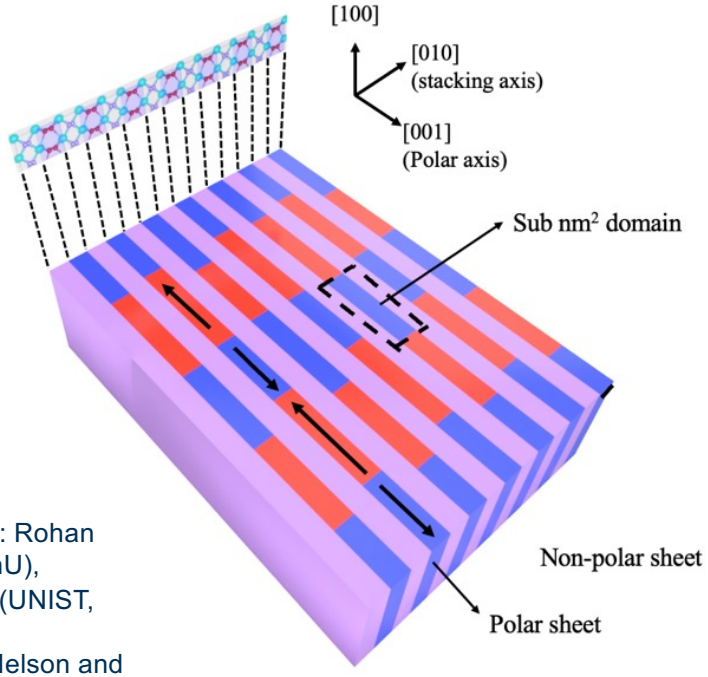
The flat transverse optical band weakens coupling between neighboring polar sheets

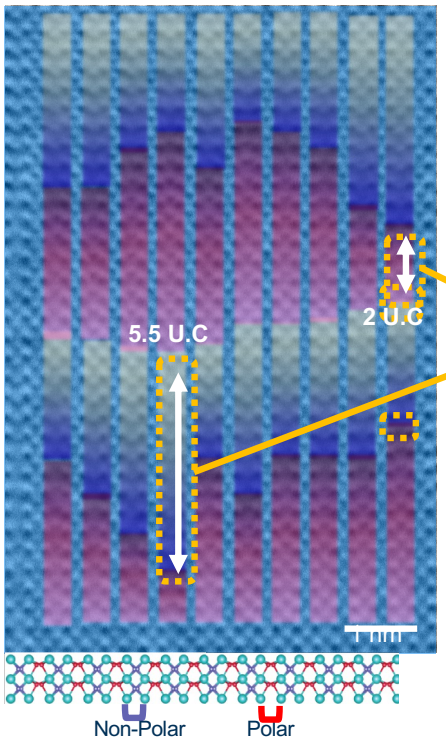


Jun Hee Lee and colleagues
H.-J. Lee et al., Science 369, 1343–1347 (2020)

DOMAIN CONFINEMENT

Domain confinement along the polar axis





0.27 nm
polar-sheet width

1-2.75 nm
polar-axis length

0.27-0.75 nm²
projected area

Collaborators: Rohan Mishra (WashU), Jun Hee Lee (UNIST, South Korea) Christopher Nelson and Andrew Lupini (ORNL)

Afroze, Nashrah, et al. "Sub-nm2 ferroelectric domains via charged 180 degree walls in ZrO2." *arXiv preprint arXiv:2507.18920* (2025). https://arxiv.org/abs/2507.18920

Source: Adv_Mat_polar_domain_boundary_marked.pdf, Figure 1

Lateral domain footprint

Perpendicular to the polar axis

~0.5 nm width (1 unit cell)

Along the polar axis

1–2.75 nm (2–5.5 unit cells), from atomic imaging

Lateral domain footprint: ~0.5–1.38 nm²

1 unit cell across × 2–5.5 unit cells along the polar axis
Structural footprint; independent electrical addressing remains to be shown

Domain switching and the lateral size limit

How small can a switchable domain be?

Binary local switching at 0.475 nm sampling
4 states (2 bits) in a 100 nm × 100 nm region
Equivalent region density: 20 Gb/cm²

How small can a polar domain become?

1 unit cell across × 2–5.5 unit cells along the polar axis
Lateral footprint: ~0.5–1.38 nm²
Projected bit density: ~73–200 Tb/cm² (1 bit/domain)

Domain density alone does not set achievable technology density.
Equally scalable read/write mechanisms (heads or access devices), addressing, and sensing are required.

ACKNOWLEDGMENTS


Prasanna Venkatesan


Lance Fernandes

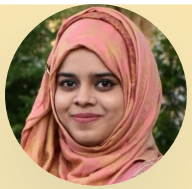

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